# COMSATS UNIVERSITY ISLAMABAD
# LAHORE CAMPUS

# ASSIGNMENT 3

## Project Planning and Management

**Project Title**

**IoT-Based Multi-Tunnel, Crop-Stage-Aware Monitoring and Decision-Support System for Strawberry Cultivation**

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# 1. Introduction

This assignment presents three project-planning components for the Final Year Project (FYP) titled “IoT-Based Multi-Tunnel, Crop-Stage-Aware Monitoring and Decision-Support System for Strawberry Cultivation.” The proposed system is intended to monitor environmental conditions across multiple strawberry cultivation tunnels, collect and transmit sensor readings through a central gateway, process the data, assess environmental risks, and provide crop-stage-aware recommendations through a farmer dashboard. A two-stage disease-response pipeline is also included: environmental risk alerts can prompt the farmer to upload a plant image for AI-based classification.

The Work Breakdown Structure (WBS) organizes the project scope into manageable work packages. The Organizational Breakdown Structure (OBS) presents a hypothetical company-style hierarchy for managing and delivering the project. The communication flow defines how instructions, progress updates, technical information, issues, and approvals move between the project team, company leadership, supervisors, and intended users.

# 2. Work Breakdown Structure (WBS)

## 2.1 Definition and Purpose

A Work Breakdown Structure is a hierarchical decomposition of the total project scope into smaller deliverables and work packages. It clarifies what must be produced, supports task assignment and estimation, and helps the team track progress. The WBS below follows the proposed FYP scope. Optional features are separated from the core deliverables so that the main project remains achievable within the available time and resources.

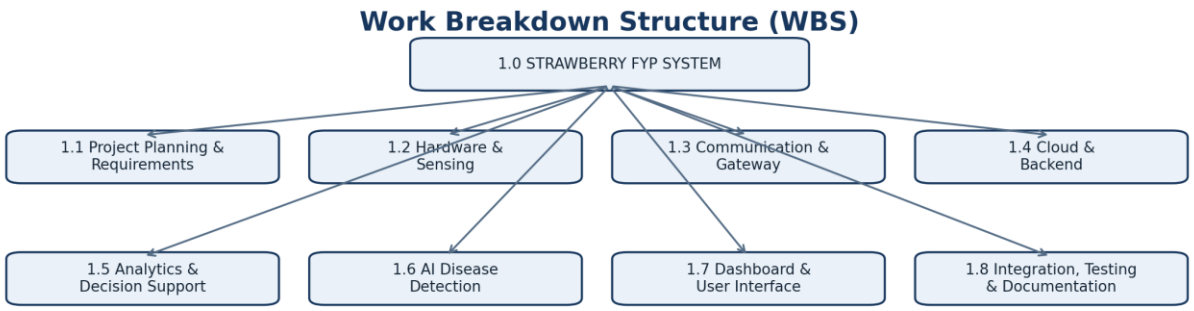



*Figure 1. Work Breakdown Structure for the Strawberry FYP*

### 2.2 WBS Dictionary / Work Package Description

| WBS Code | Work Package | Description / Activities | Expected Deliverable |
|---|---|---|---|
| 1.1 | Project Planning and Requirements | Review related work; confirm scope, functional requirements, constraints, risks, milestones and evaluation plan. | Requirements specification and project plan |
| 1.2 | Hardware and Sensing | Select and integrate microcontroller-based sensor nodes; measure temperature, relative humidity and soil moisture; calibrate sensors; identify each tunnel. | Calibrated tunnel sensor nodes |
| 1.3 | Communication and Gateway | Develop node firmware and ESP-NOW communication; implement tunnel IDs, timestamps, packet checks, gateway reception and temporary buffering. | Tested node-to-gateway data path |
| 1.4 | Cloud and Backend | Receive gateway data; validate fields and values; store timestamped tunnel records; provide backend services/APIs and historical data access. | Working backend and database |
| 1.5 | Analytics and Decision Support | Calculate VPD and trends; interpret soil-moisture status; apply crop-stage-aware rules; generate explainable tunnel-specific irrigation, ventilation and environmental-risk advisories. | Risk indicators and explainable advisories |
| 1.6 | AI Disease Detection | Build the two-stage pipeline: environmental risk prompts photo upload; image classifier identifies Botrytis, Powdery Mildew, Crown Rot or Healthy and presents relevant treatment guidance. | Integrated image-classification module |
| 1.7 | Dashboard and User Interface | Display live tunnel readings, historical trends, tunnel comparison, crop-stage selection, system status, alerts and recommendations. | Farmer-facing dashboard |
| 1.8 | Integration, Testing and Documentation | Integrate hardware and software; test sensor accuracy, communication reliability, response time, scalability, power, usability and cost; prepare final report and demonstration. | Evaluated prototype, documentation and final demonstration |
| 1.8.1 | Optional Extensions | If time, data and resources allow, investigate historical replay, optional leaf-wetness sensing, or selective automatic irrigation/ventilation with safeguards. | Optional extension only; not required for core completion |

### 2.3 WBS Management Notes

- The WBS includes the full core scope described in the FYP proposal: multi-tunnel sensing, gateway communication, cloud processing, risk assessment, crop-stage-aware decision support, AI-based disease classification, dashboard, and evaluation.
- Each work package should be further divided into individual tasks when the team prepares the detailed schedule.
- Optional automatic irrigation/ventilation and historical replay should not delay delivery of the core monitoring and advisory system.
- Completion should be assessed against measurable acceptance criteria, including sensor accuracy, packet delivery, end-to-end response time, classifier performance, usability, power and cost.

## 3. Organizational Breakdown Structure (OBS)

### 3.1 Definition and Purpose

An Organizational Breakdown Structure represents the project organization by showing roles, teams, and reporting relationships. For this assignment, the project is represented as a hypothetical technology company developing an agricultural IoT product. The CEO, co-founders, and managers below are proposed organizational roles for planning purposes; they are not claims that a registered company or these formal job titles already exist. The academic FYP supervisors remain the project's university supervisory stakeholders.

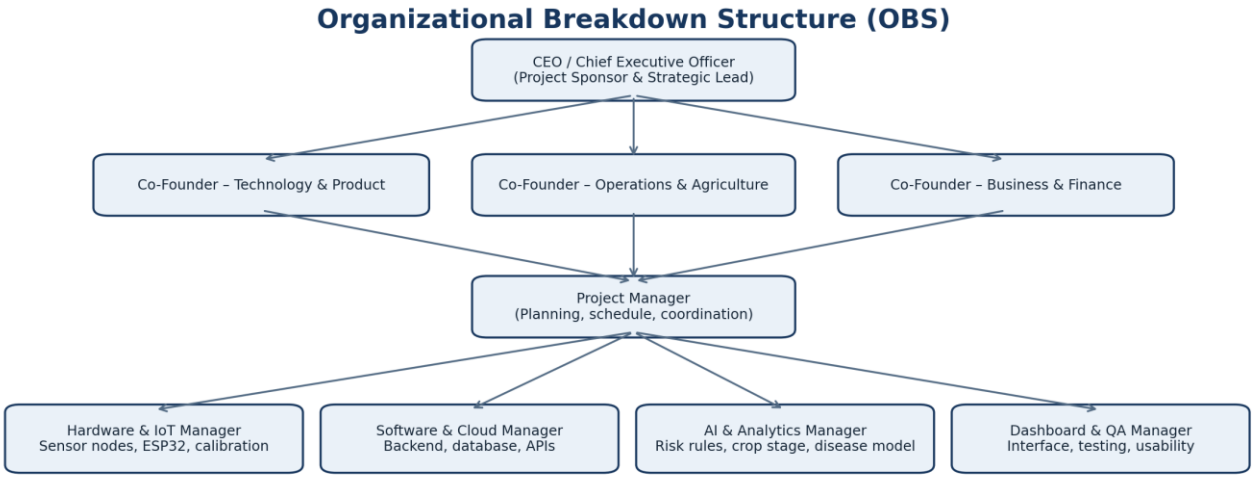



*Figure 2. Company-Style Organizational Breakdown Structure (proposed)*

## 3.2 Roles and Responsibilities

| Role / Stakeholder | Main Responsibilities |
|---|---|
| CEO / Chief Executive Officer | Overall strategic direction, project sponsorship, resource priorities, major decisions and high-level review. |
| Co-Founder – Technology and Product | Oversees technical architecture, integration decisions, product quality and alignment between hardware and software. |
| Co-Founder – Operations and Agriculture | Coordinates practical cultivation requirements, farmer needs, field access, operational constraints and validation context. |
| Co-Founder – Business and Finance | Tracks budget, equipment procurement, cost per tunnel, resource use and potential product viability. |
| Project Manager | Maintains the project plan, schedule, milestones, risk register, meetings, task allocation and progress reporting. |
| Hardware and IoT Manager | Leads sensor selection, ESP32-based node implementation, calibration, power planning, ESP-NOW communication and gateway integration. |
| Software and Cloud Manager | Leads backend services, data validation, database, APIs, cloud ingestion and historical records. |
| AI and Analytics Manager | Leads environmental indicators, risk rules, crop-stage-aware recommendations and the AI disease image-classification pipeline. |
| Dashboard and Quality Assurance Manager | Leads dashboard usability, live displays, alerts, system integration test planning, defect tracking and acceptance evidence. |
| FYP Supervisors | Dr. Babar Ali (Supervisor) and Dr. Farooq-e-Azam (Co-Supervisor) provide academic guidance, review progress, advise on technical/research direction and assess project milestones. |
| FYP Development Team | Abdul Mateen (FA23-BCE-004), Hassan Zafar (FA23-BCE-037), and Muhammad Usman (FA23-BCE-073) collaboratively develop and evaluate the system. Specific manager assignments can be confirmed by the team. |

## 3.3 Organizational Notes

- The company-style hierarchy is a planning model created to satisfy the assignment's organizational view; it does not imply that a company has been legally formed.
- The CEO and co-founders provide strategic oversight, while the project manager coordinates day-to-day delivery and managers lead functional workstreams.
- The three FYP students form the actual academic project team. They may hold more than one functional role depending on skills and workload.
- University supervisors are shown as academic stakeholders rather than employees reporting to the hypothetical company CEO.

# 4. Project Communication Flow

## 4.1 Purpose

Good communication keeps everyone working from the same plan. This section explains who talks to whom, how often, and what to do when something goes wrong. It uses the same roles shown in the OBS in Section 3, so every person can see exactly where they fit.

Messages travel in three directions: down from leadership to the teams, up from the teams to leadership, and sideways between teams that depend on each other. Two outside groups, the FYP supervisors and the farmer, connect to the team at specific points.

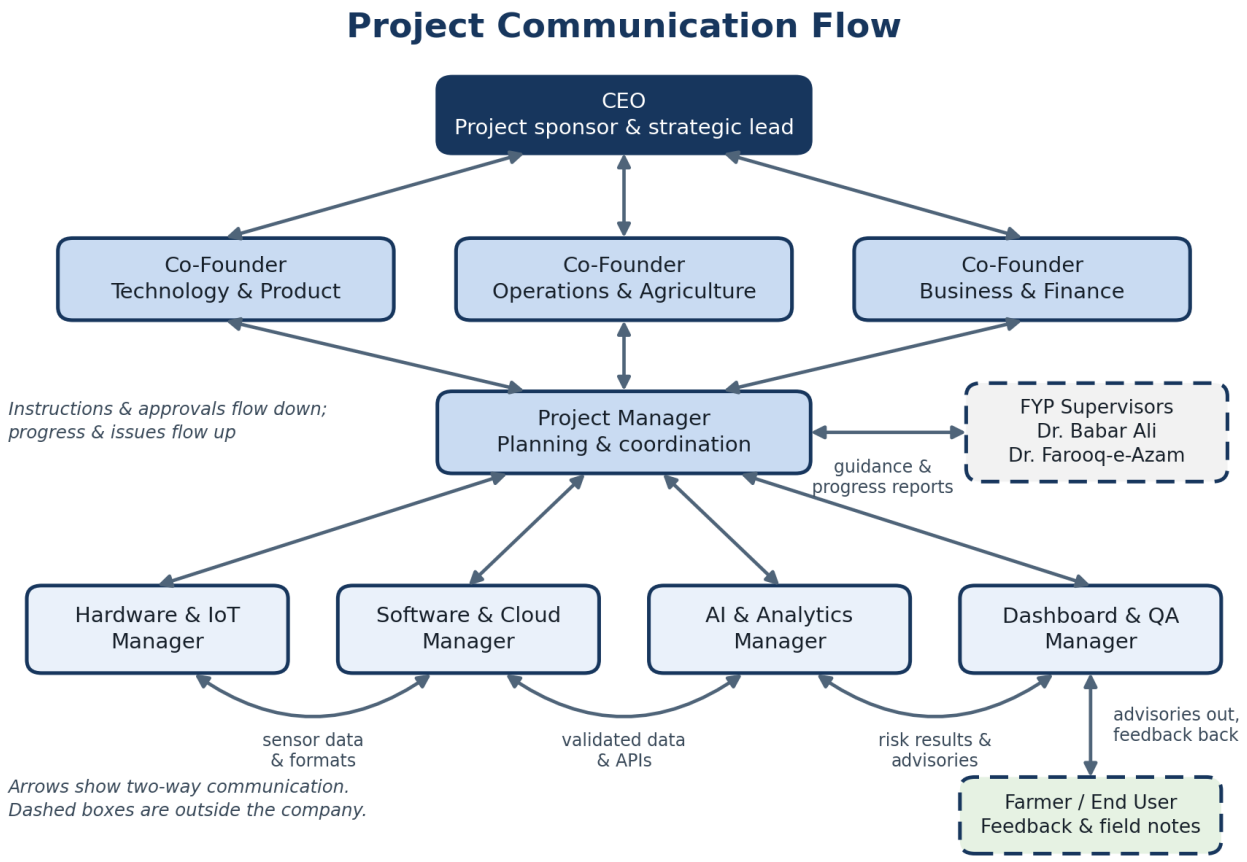



*Figure 3. Communication Flow for the Strawberry FYP*

## 4.2 How Information Moves

- **Downward.** The CEO and co-founders set priorities and approve the scope. The Project Manager turns these into clear tasks and deadlines for the four managers.
- **Upward.** Each manager reports progress, problems and test results to the Project Manager. The Project Manager passes a short summary to the co-founders and CEO. Anything that needs a decision travels up the same way.
- **Sideways.** The four managers talk directly to each other because their work connects. Hardware sends sensor data to software, software passes clean data to analytics, and analytics hands its advisories to the dashboard.
- **Outside the company.** The FYP supervisors review and guide the project through the Project Manager. The farmer gives feedback on the dashboard, mainly through the Dashboard and QA Manager.

## 4.3 Who Communicates with Whom

The list below follows the OBS from top to bottom. Because the three FYP students may hold more than one manager role, many of these conversations will happen between the same few people, but the same rules still apply.

**CEO.** Talks mainly with the three co-founders. Sets the overall direction, receives short progress summaries and makes the final call on major decisions such as scope changes or extra resources.

**Co-Founders.** Each co-founder follows the part of the project that matches their area. The Technology and Product co-founder handles hardware and software questions, the Operations and Agriculture co-founder handles farmer needs and field access, and the Business and Finance co-founder handles budget and equipment. They give guidance and approvals to the Project Manager and take serious issues to the CEO.

**Project Manager.** The central point of contact. Passes priorities down to the managers, collects their weekly updates, keeps the task tracker and risk list, and is the link to the FYP supervisors.

**Hardware and IoT Manager.** Shares sensor data formats, tunnel IDs and packet details with the Software and Cloud Manager. Reports node or gateway problems to the Project Manager straight away.

**Software and Cloud Manager.** Receives data from the hardware side, checks and stores it, and shares API details and clean data with the AI and Analytics Manager and the dashboard team.

**AI and Analytics Manager.** Uses the stored data to work out risk levels and advisories, and passes these, along with the disease-classification results, to the Dashboard and QA Manager. Tells the software team what data is needed.

**Dashboard and QA Manager.** Shows advisories and alerts to the farmer, collects the farmer's feedback, runs integration tests, and sends bug reports back to whichever team owns the problem.

**FYP Supervisors.** Dr. Babar Ali and Dr. Farooq-e-Azam sit outside the company structure. They review progress at agreed meetings, advise on technical direction and give feedback on the report and demonstration.

**Farmer / End User.** Tries the prototype during demonstrations and usability testing, and tells the team what is unclear, useful or impractical in the field.

## 4.4 Meetings and Updates

- **Weekly team meeting.** Everyone meets, led by the Project Manager, to go over progress, next tasks and anything blocking the work.
- **Weekly manager update.** Each manager sends a short status note to the Project Manager through the shared tracker or group chat, and again whenever a milestone is reached.
- **Technical catch-ups.** Held as needed. The managers involved agree on data formats, interfaces and fixes, and log the outcome in the issue tracker.
- **Supervisor reviews.** The Project Manager and team meet the supervisors on an agreed schedule and send a short written update beforehand.
- **Farmer feedback sessions.** Held during prototype demonstrations and usability testing, using a short feedback form.
- **Milestone approvals.** The Project Manager presents finished deliverables to the co-founders and CEO for sign-off. The team also submits key deliverables to the supervisors for academic review.

### 4.5 When Something Goes Wrong

Problems such as a hardware failure, lost data, a major delay or a change in scope are handled in the same order every time.

1. **Write it down.** The person who spots the problem notes what happened, what it affects and anything already tried.
2. **Tell the right people quickly.** Report it to their manager and the Project Manager by message or call, followed by a short written note.
3. **Assign an owner.** The Project Manager decides who will fix it and sets a target date.
4. **Escalate if needed.** If the issue affects scope, budget, a key milestone, data integrity or safe operation, the Project Manager takes it to the relevant co-founder or the CEO, and to the FYP supervisor if academic guidance is needed.
5. **Record the outcome.** The decision and follow-up actions are saved in the meeting notes or project tracker.

### 4.6 Ground Rules

- Use one shared tracker to record each task's owner, due date, status, blockers and proof of completion.
- Write down major decisions, scope changes and supervisor feedback after every meeting.
- Always include the tunnel ID and timestamp in technical messages, test logs and fault reports so data can be traced.
- Raise risks early instead of waiting for the next meeting when something important is blocked.
- Share only validated test results, and clearly label features as core or optional.

## 5. Conclusion

The WBS, OBS and communication flow provide complementary views of the strawberry monitoring and decision-support FYP. The WBS defines the work and expected deliverables, from sensor nodes and gateway communication to cloud processing, crop-stage-aware advisories, AI disease classification, dashboard development and system evaluation. The OBS establishes a company-style reporting model while retaining the actual FYP students and university supervisors as project stakeholders. The communication plan explains how teams coordinate technical dependencies, report progress, receive feedback and escalate issues. Together, these structures support clearer accountability, more controlled project execution and delivery of the core system within the planned scope.